JAN. FEB. MAR. APR. MAY JUN. JUL. AUG. SEP. OCT. NOV. DEC.

# CSA

Virtual Event

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# AI Cyber Threats: By the Numbers

PRESENTED BY

Dr. Christopher Porter, PhD

# GTI Nation Grade Collection + Analysis w/ Gemini AI

### Incident Response Intelligence

- 1000+ engagements Per Year
- ~ 60 Countries

### Operational Intelligence

- Four Globally Positioned Security Operations Centers

### VirusTotal Intelligence

- 20+ Years + 191 Countries submitting files

### Adversary Intelligence

- 350 Researchers – 30+ languages

### Machine Intelligence

- Defend 2.5B Android devices daily.
- Defend 4B+Gmail and Chrome Users

### Human Expertise

- 500+ personnel in the intelligence business unit
- 400+ Red Team exercises per year



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# Long-Term

## Attack

- Lower cost/barrier to entry
- More reliance on insider threats

## Defense

- Speed
- Scale
- Fragility?

## HR

- Judgment >> implementation
- Key dilemma: how to train and retain to create those senior analysts with judgment
- Analysis largely becomes a skill not a profession
- Cost of prediction down

# 30% increase in fuzzer coverage for critical open source

```
extern "C" int LLVMFuzzerTestOneInput(const uint8_t* data, size_t size) {
  std::string data_string(reinterpret_cast<const char*>(data), size);
  tinyxml2::XMLDocument doc;
  doc.Parse(data_string.c_str());

  return 0;
}
```

**+2.04%** coverage

```
tinyxml2::XMLNode* node =
doc.FirstChild();
  if (node != nullptr) {
    node->ShallowEqual(node);
}
```

```
XMLNode *node = doc.FirstChild();
if (node) {
  XMLElement *element = node->ToElement();
  if (element) {
    element->ShallowClone( &doc );
  }
}
```

**+2.07%** coverage

: **+11.12%** coverage

```
tinyxml2::XMLPrinter printer;
doc.Print(&printer);
```

**+10.25%** coverage

```
doc.SaveFile("out.xml",true);
```

**+3.54%** coverage

```
XMLElement *root = doc.FirstChildElement( "root" );
if (root == nullptr) {
  root = doc.NewElement( "root" );
  doc.InsertFirstChild( root );
}
XMLElement *child = doc.NewElement( "child" );
root->InsertEndChild( child );
child->SetAttribute( "name", "foo" );
child->SetAttribute( "value", 1 );
```

# Medium-Term

## Attack

- Broader array of actors finding 0 days
- Open-weight models
- AI tool-use

## Defense

- AI code review moves 0 days from memory allocation to logic/stateful errors
- Faster patching
- Automated response to intrusion attempts

## HR

- Significant upheaval in mid-ranks from AI-savvy and not
- Hiring experts in anomalous human behavior, not just anomalous technical/network behavior
- Fusion of human expertise and autonomous agents

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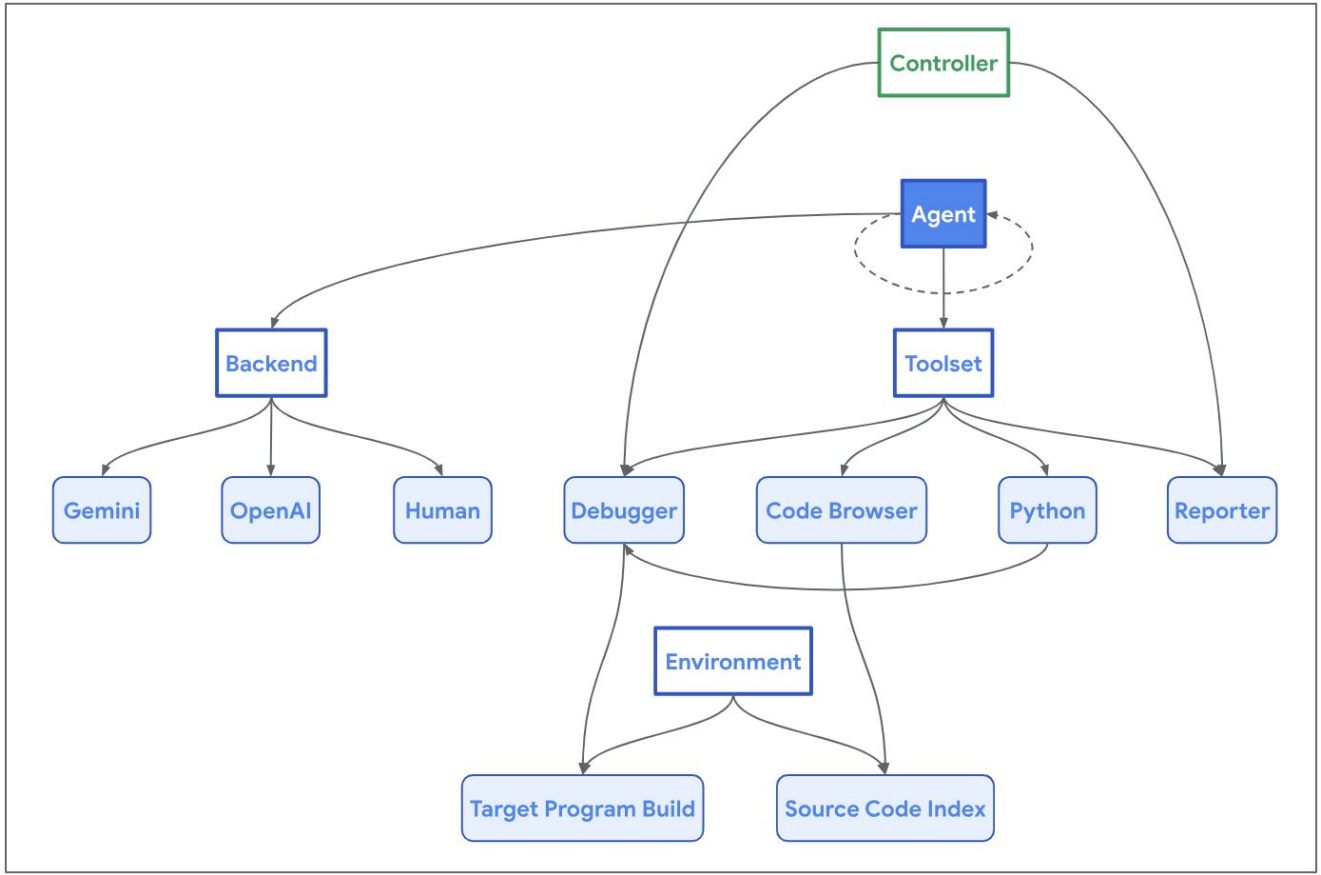
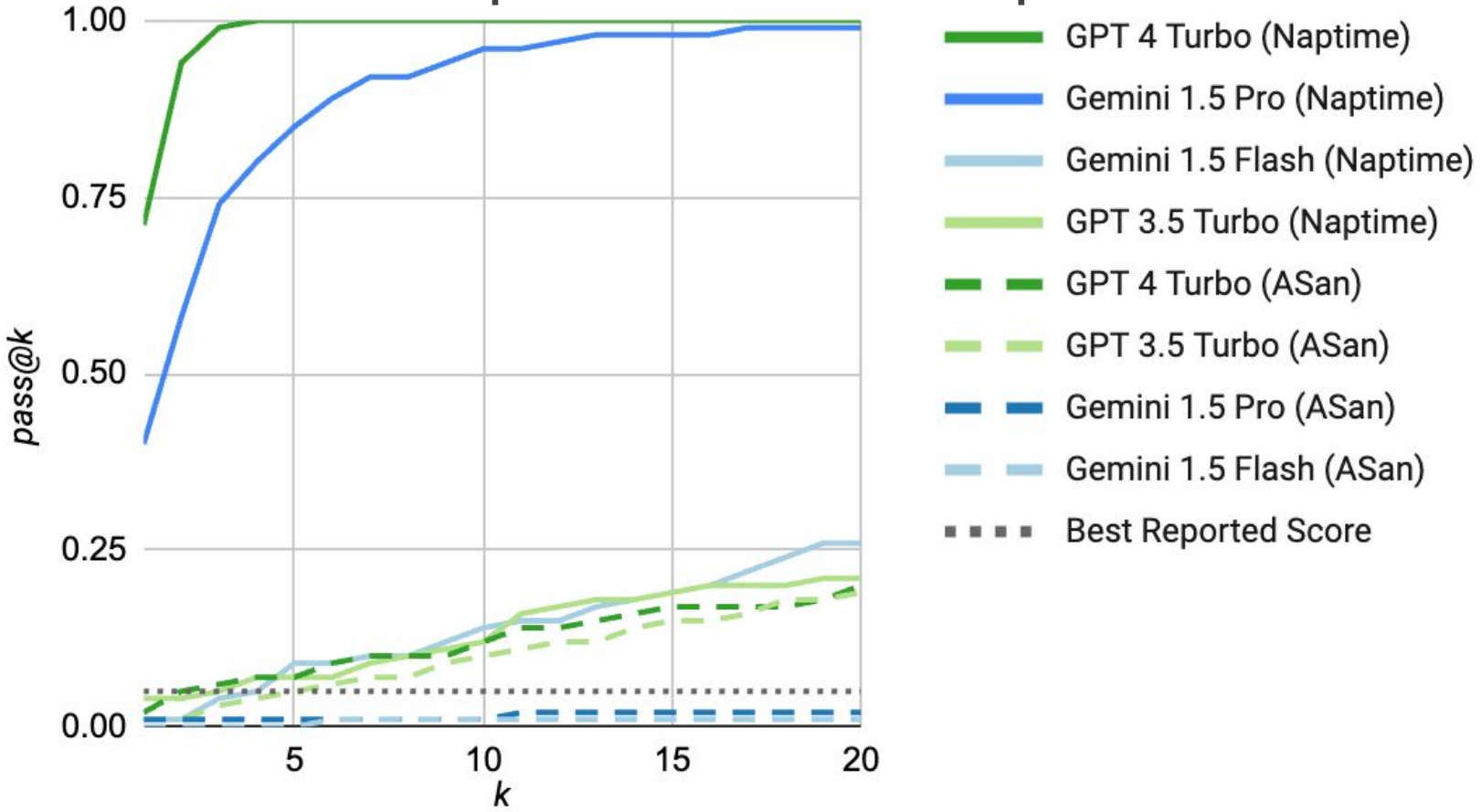
**By combining our AI and threat intelligence capabilities, Google predicted and foiled an imminent attack.**

**We believe this is a first for an AI agent.**

# Naptime Architecture: Recreating Security Researcher Environment and Tools



# Buffer Overflow Attempts with and without Naptime Architecture



# Vulnerability fixing

We harnessed our Gemini model to successfully fix 15% of bugs discovered by our sanitizer tools during testing, resulting in hundreds of bugs patched. We expect this success rate to continually improve and anticipate that LLMs can be used to fix bugs in various languages across the software development lifecycle.

# Short-Term

## Attack

- Automation of attack phases → more exploited victims when N-days are used
- Translation -> broader array of threat groups as regional powers go global

## Defense

- AI > CTI analysts at drafting routine reports and not missing key intel
- Agentic analysts

## HR

- 10X output
- Authorship becomes a more independent process
- Coordinated decision-making within an organization will use LLMs as the base layer, not bureaucratic coordination

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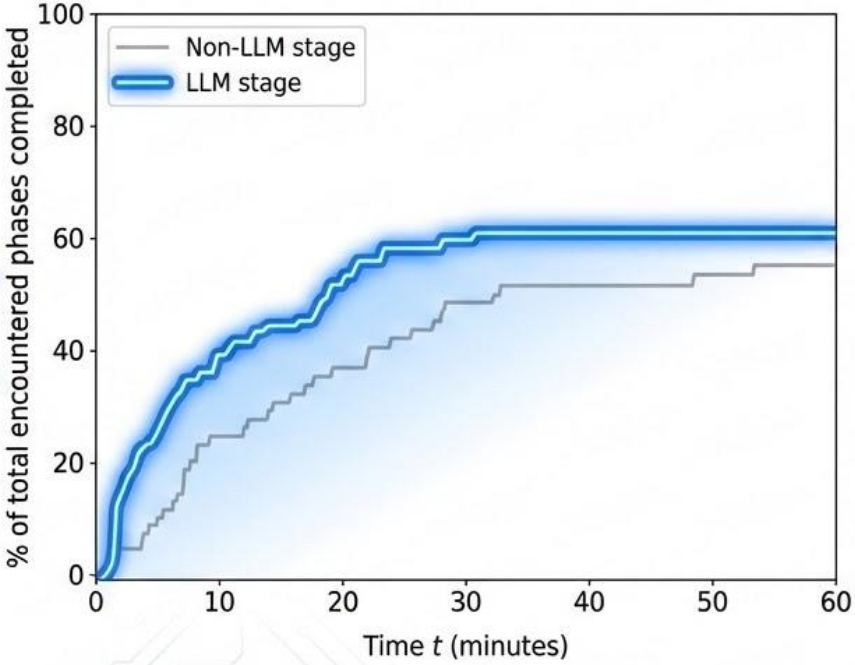
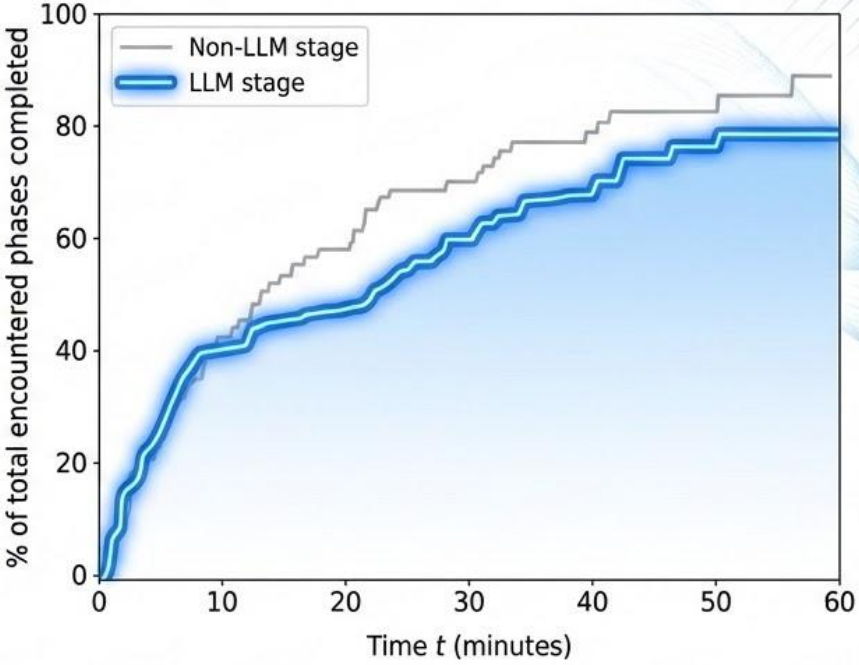
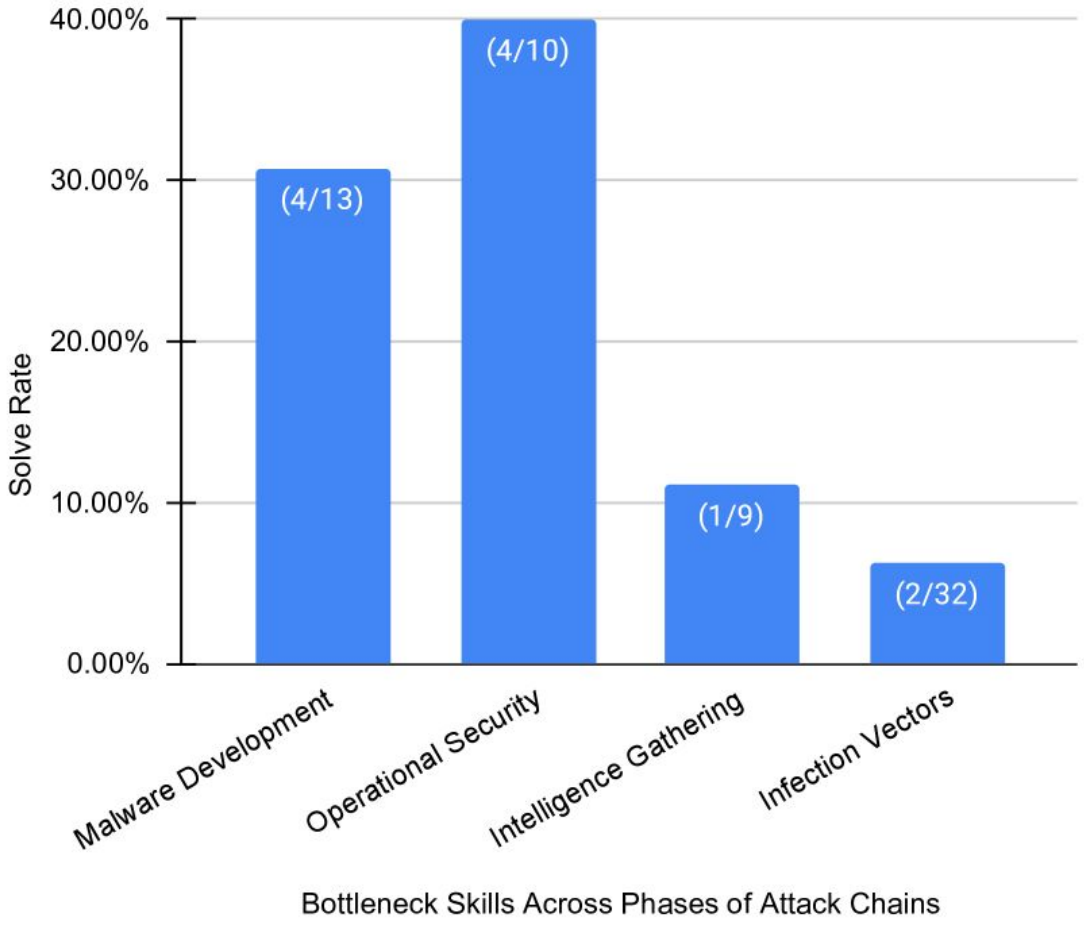
# Phase Completion Rate: Novice vs. Expert

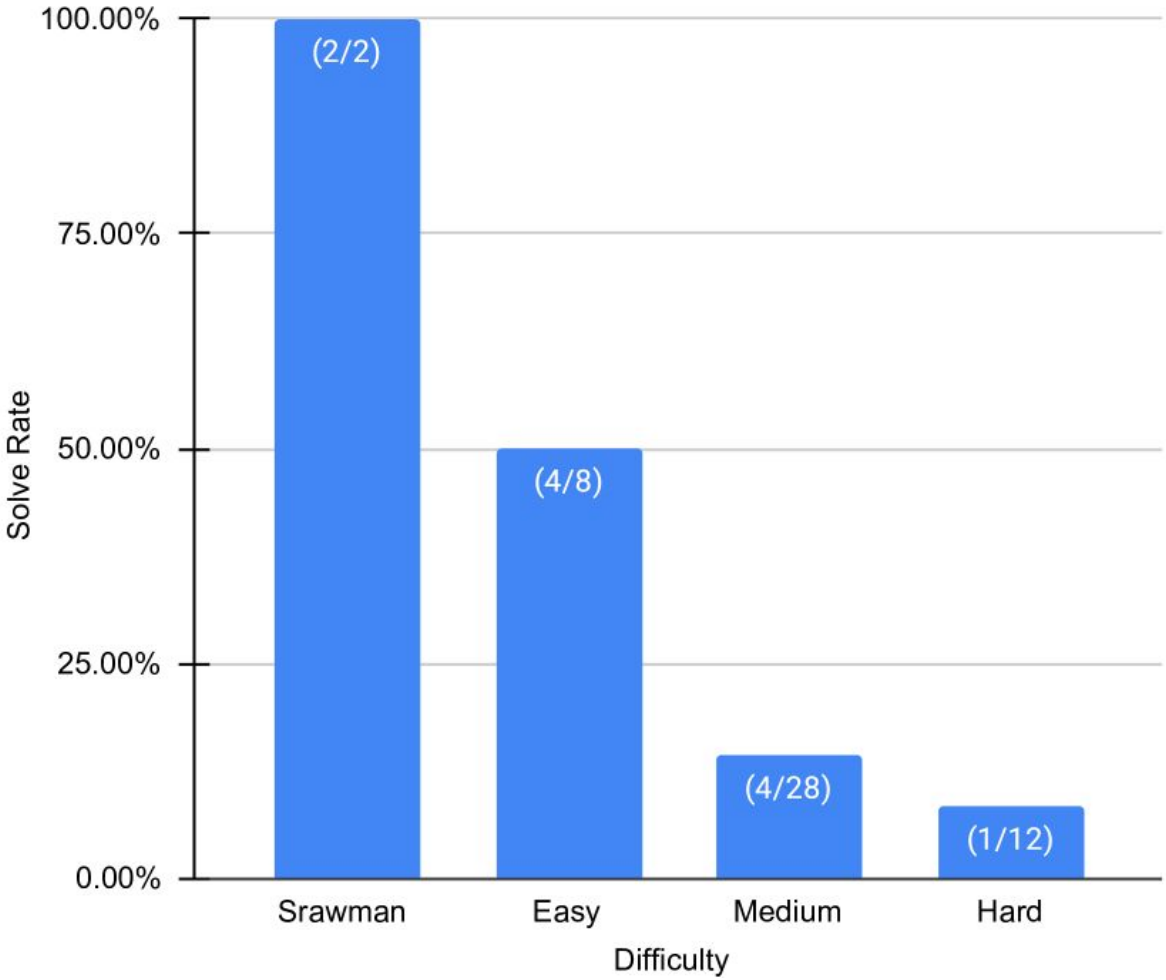
## Novice phase completion rate

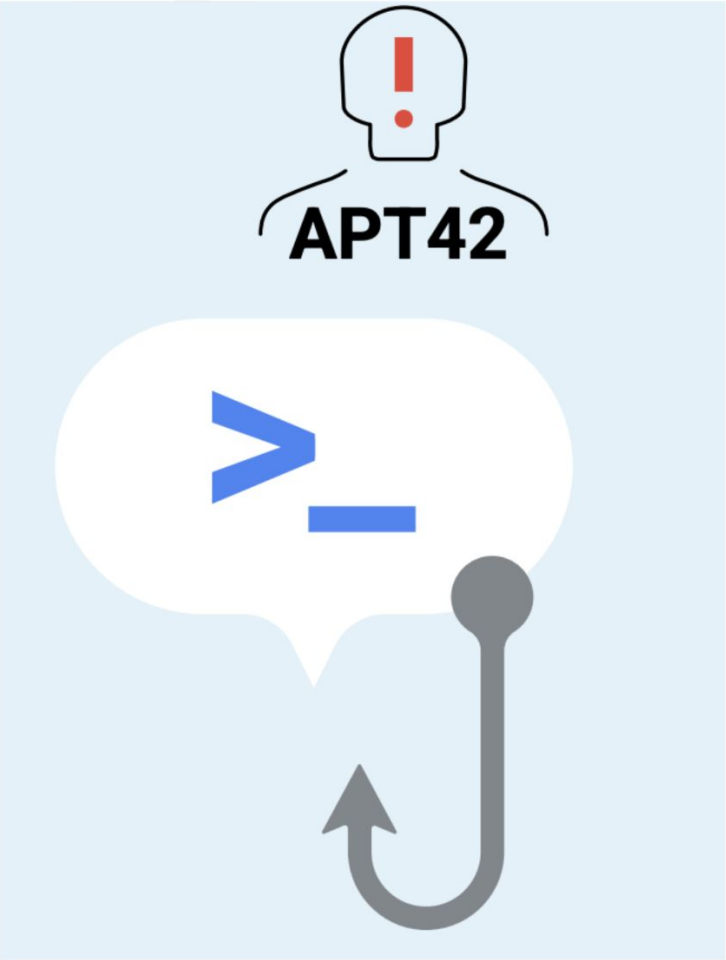



## Expert phase completion rate

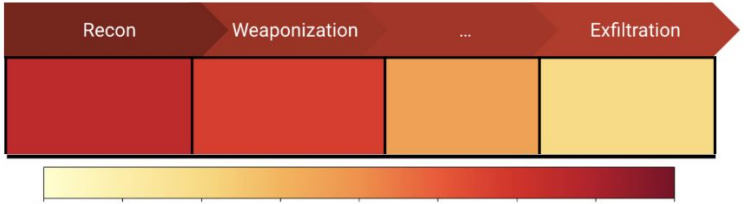



40.00%
30.00%
20.00%
10.00%
0.00%
Solve Rate
(4/13)
(4/10)
(1/9)
(2/32)
Malware Development
Operational Security
Intelligence Gathering
Infection Vectors
Bottleneck Skills Across Phases of Attack Chains

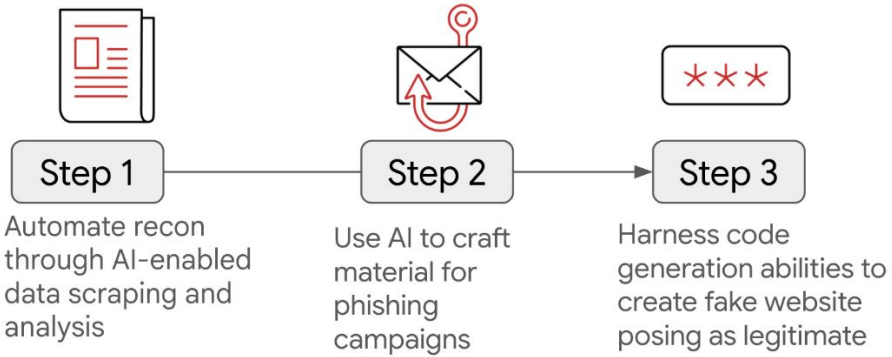

100.00%
75.00%
50.00%
25.00%
0.00%
Solve Rate
(2/2)
(4/8)
(4/28)
(1/12)
Srawman
Easy
Medium
Hard
Difficulty




**APT42, an Iranian state-sponsored cyber espionage actor, uses enhanced social engineering schemes to gain access to victim networks, including cloud environments. Targets Western and Middle Eastern NGOs, media organizations, academia, legal services and activists.**




## AI-enabled cost reduction



# Iranian APT Actors the Heaviest Users of Gemini.

We observed a broad scope of research and use cases, that reflected strategic Iranian interests

Over 30% of Iranian APT actors' Gemini use was linked to **APT42** who used Gemini to:

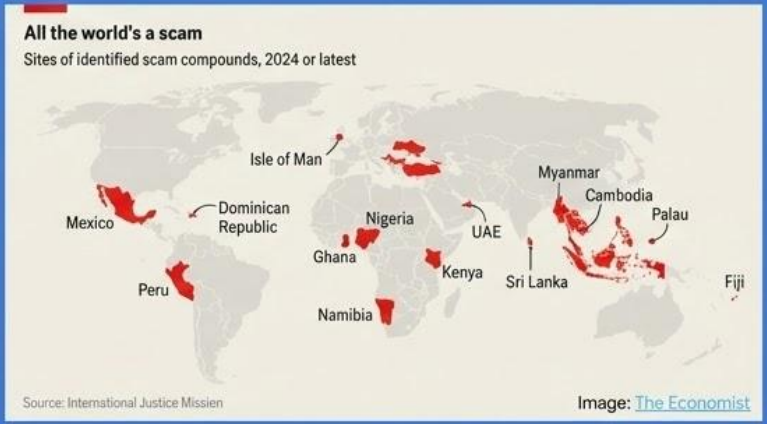
- Craft phishing campaigns,
- Conduct reconnaissance into individual policy and defense experts
- Research publicly known vulnerabilities.

# Scams: Adopting AI, enabled by cybercrime

## Growing Issue

- Online fraud global proceeds estimated to exceede to $500bn - $1n per year
- Pig butchering in 2024+ 40% YoY revenue; +210% number of deposits

## Transnational Cybercrime

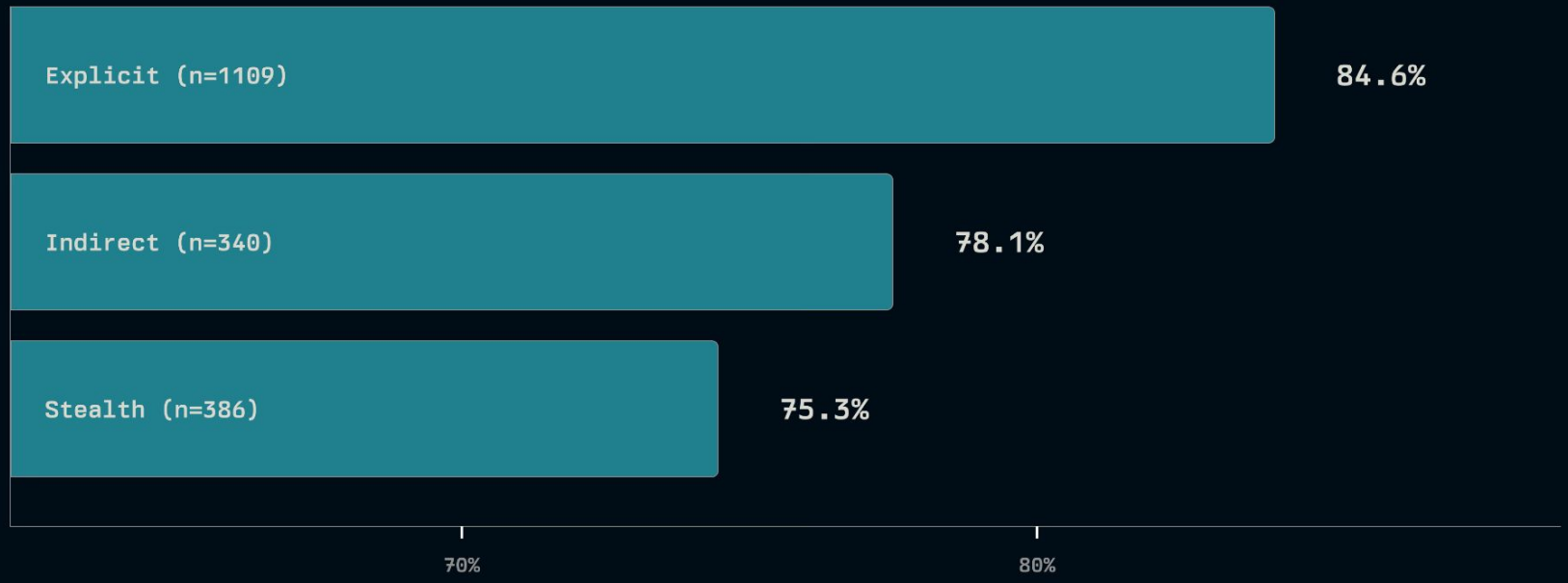



- Commodity scam activity evolving into
- Major crackdown in China has shifted scaled scam activity to other geos
- Corruption nearing state capture in some geos (Cambodia, the Philippines)

## Cyber Crime as a Scam Enabler

- Use infostealers to harvest data at acotunt enabliers (e.g., messages, browsing history)
- Sell data to scammers and sells subscription services

## AVERAGE BALANCED ACCURACY



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# AI in Intelligence Analysis



Google Cloud

# Malware Analysis

Gemini can understand how code behaves in a deep way to spot new threats, including ones never seen before.

**27**

**Seconds**
to produce a complete malware analysis report

# AI to Write Intelligence Reports

## Data

Analyst Research

Finished Intelligence Corpus

Internal Threat Databases

Open Source Information

Raw Intelligence Sources

Etc.

## AI Tooling

Gemini

## Draft Research & Reports

**Intel Overview: BlackMatter Ransomware and Associated Threat Actors**

**Date:** 2025-07-16
**Focus:** Agriculture, Food Supply Chain
**Timeframe:** September 2021

**Bottom Line Up Front (BLUF):**
Recent analysis of a cyber attack against the NEW Cooperative agricultural group indicates the involvement of the BlackMatter ransomware. This activity is tracked by GTI under the financially motivated threat cluster UNC2628, which operates as a ransomware affiliate, and is also associated with the broader threat cluster UNC2900. (21-00020678, threat-actor--31a68f18-a869-5062-8164-8417769a4161, threat-actor--df3ee9b4-fae2-5c18-83ae-0491cc3f4394)

**Key Highlights & Supporting Details:**

- **Targeting of US Agriculture:** In September 2021, a ransomware attack impacted the operations of NEW Cooperative, an agricultural business based in Iowa. The actors claimed to have exfiltrated financial data, human resources information, and source code for the cooperative's proprietary soil-mapping technology. (21-00020678)
- **Known Aliases:** The threat actor UNC2900, which has deployed BlackMatter ransomware, is also known by several public names including Lockean (ANSSI), Blackbasta (Fortinet), Blackcat (Trend Micro), and Tunnel Spider (CrowdStrike). (threat-actor--df3ee9b4-fae2-5c18-83ae-0491cc3f4394)
- **Modus Operandi:** The threat cluster UNC2628 is a financially motivated ransomware affiliate known to use stolen credentials or password guessing for initial access. They deploy tools like BEACON and ANYDESK for persistence and use various utilities for internal reconnaissance and data exfiltration before deploying ransomware. (threat-actor--31a68f18-a869-5062-8164-8417769a4161)
- **Multi-Ransomware Operations:** The broader threat cluster UNC2900 has been observed deploying a variety of ransomware families in their operations, including CONTI, SODINOKIBI (REvil), ROYALLOCKER, and BLACKMATTER, indicating a versatile and opportunistic approach to their extortion campaigns. (threat-actor--df3ee9b4-fae2-5c18-83ae-0491cc3f4394)

**Some reports now taking 95% less time to create**

# LLM are able to help incident teams write incident summaries 51% faster

https://security.googleblog.com/2024/04/accelerating-incident-response-using.html

# Thank you!